Replicating xRFM: Accurate, Scalable, and Interpretable Feature Learning Models for Tabular Data +bonus

COMP9417 Group Project

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1 Introduction

Tabular data remains one of the most common data modalities in applied machine learning, appearing in domains such as materials science, healthcare, e-commerce, and human-resource analytics. Despite the rapid progress of deep learning on images, language, and sequential data, tabular learning is still difficult because useful signals are often sparse, heterogeneous, and distributed across mixed numerical and categorical variables. Prior benchmarking studies show that gradient-boosted decision trees (GBDTs), including XGBoost and LightGBM, remain difficult to outperform on many typical tabular datasets because their axis-aligned splits, robustness to feature scale, and ability to model non-linear interactions match the structure of many tabular problems [Chen and Guestrin, 2016, Ke et al., 2017, Grinsztajn et al., 2022, Holzmüller et al., 2024].

This project replicates and evaluates xRFM, a recent feature learning approach for tabular data based on the average gradient outer product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. Instead of relying only on fixed input features, xRFM uses gradient-based information from a learned prediction function to estimate directions in the input space that are most relevant to the target. This produces both a predictive model and an interpretable feature-relevance signal. The central question of our replication is therefore whether AGOP-driven feature learning can offer a competitive alternative to strong tabular baselines, especially when compared with XGBoost, LightGBM, and a multilayer perceptron (MLP).

We evaluate all models under a unified experimental pipeline across five public tabular datasets: Diamonds, Superconductivity, Online Shoppers Purchasing Intention, Stroke Prediction, and HR Attrition. These datasets cover regression and binary classification, large- and small-sample regimes, low- and high-dimensional feature spaces, and mixed feature types. The report is organised as follows. Section 2 describes the xRFM algorithm, the AGOP interpretation, dataset preprocessing, evaluation metrics, and model configurations. Section 3 compares predictive performance, training efficiency, scalability, and interpretability. Section 4 discusses when xRFM appears useful and when it is clearly dominated by tree-based methods. Section 5 summarises the main findings and future directions.

Overall, our results show that tree-based models remain the strongest general-purpose baselines for these tabular tasks. xRFM is competitive on selected small-scale classification settings, particularly HR Attrition, but underperforms substantially on large-scale regression such as Diamonds. This suggests that AGOP-based feature learning is promising as an interpretable modelling strategy, but it requires careful tuning and may not yet replace highly optimised GBDT pipelines for general tabular prediction.

2 Methodology

2.1 xRFM: Algorithm and Theoretical Background

xRFM learns task-relevant feature directions by iteratively estimating the Average Gradient Outer Product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. For a trained model $f : \mathbb{R}^d \to \mathbb{R}$, the AGOP is

$$G_f = \mathbb{E}_x\left[\nabla f(x) \nabla f(x)^\top\right] \in \mathbb{R}^{d\times d}, \tag{1}$$

where large diagonal entries of G_f indicate features to which f is most sensitive. Unlike PCA (unsupervised input-variance directions [Pearson, 1901]), mutual information (univariate target dependence [Kraskov et al., 2004]), or permutation importance (post-hoc performance drop [Fisher et al., 2019]), AGOP captures supervised, multivariate sensitivity arising directly

from the model's gradients. The diagonal entries provide coordinate-wise feature importance; the leading eigenvectors reveal task-relevant linear combinations of inputs.

The xRFM training pipeline iterates: (1) fit a kernel/random-feature model on the current feature space; (2) estimate G_f from training gradients; (3) reweight the feature space by G_f to focus on predictive directions; repeat for a fixed number of iterations. Inference routes a new point x^* through the learned adaptive tree to the appropriate leaf model. We use diagonal AGOP (`use_diag=True`) throughout to reduce memory cost from $O(d^2)$ to $O(d)$ while preserving feature-level interpretability.

2.2 Experimental Design

2.2.1 Datasets

We evaluate on five publicly available tabular datasets (Table 1) covering two regression and three binary classification tasks, with n ranging from 882 to 32,365 training examples, d up to 81 features, and both pure-numerical and mixed feature types [Hamidieh, 2018, Sakar et al., 2019, fedesoriano, 2021].

Table 1: Dataset statistics. n_{train}, n_{val}, and n_{test} denote the sizes of the training, validation, and test splits respectively; d is the number of features after preprocessing.

Dataset	Task	Target	n_{train}	n_{val}	n_{test}	d
Diamonds	Regression	price	32,365	10,789	10,789	26
Superconductivity	Regression	critical_temp	12,757	4,253	4,253	81
Online Shoppers	Classification	Revenue	7,398	2,466	2,466	27
Stroke	Classification	stroke	3,066	1,022	1,022	21
HR Attrition	Classification	Attrition	882	294	294	51

2.2.2 Data Preprocessing and Splitting Strategy

To maintain fairness and consistency across all experiments, all random seeds were fixed at 42 for reproducibility. For each dataset, we applied a strict 60/20/20 split for the training, validation, and test sets. Missing values were imputed using the column-wise median for numerical features and the constant `Missing` for categorical features. Numerical features were standardised using z-score normalisation, and categorical features were transformed using one-hot encoding. The processed features were concatenated into a single numerical array before being passed to all models, ensuring compatibility with xRFM's AGOP computations and with the baseline models.

2.2.3 Evaluation Metrics

Regression tasks are evaluated by RMSE (lower is better); classification tasks by accuracy and AUC-ROC, where AUC is preferred under class imbalance. We also record training time and per-sample inference time for all models. All hyperparameters are selected on the validation set; test metrics are computed once on the held-out test set.

2.2.4 xRFM Configuration

The xRFM model was implemented using the Python `xrfm` package and trained through the shared project pipeline. The training script initialises the model as `xRFM(use_diag=True, device=DEVICE)`, where `DEVICE` is set to CUDA when available and CPU otherwise. The diagonal setting was selected because it substantially reduces the memory cost of AGOP estimation while producing feature-level importance values. The project code also fixes NumPy and PyTorch

random seeds at 42, which improves reproducibility across repeated runs [Pedregosa et al., 2011, Paszke et al., 2019].

When running on GPU, the code reduces the number of xRFM fit iterations to 3 and sets `M_batch_size=128`. This setting was used as a practical compromise between computational cost and model quality. All xRFM models were fitted on the training split, monitored using the validation split, and evaluated once on the held-out test split. For classification tasks, the code used predicted class labels for accuracy and attempted to use `predict_proba` scores for AUC-ROC. For regression tasks, predictions were evaluated directly using RMSE. Feature importance was extracted from the learned model by reading the AGOP matrix or diagonal values from the internal xRFM tree model and ranking features by their AGOP-derived scores.

Table 2: xRFM configuration used in the experiments.

Component	Setting
Model	`xRFM`
AGOP mode	Diagonal AGOP (`use_diag=True`)
Device	CUDA if available; otherwise CPU
Random seed	42
GPU fit iterations	3
GPU AGOP batch size	128
Input representation	Standardised numerical features and one-hot encoded categorical features
Evaluation split	60/20/20 train/validation/test split
Regression metric	RMSE
Classification metrics	Accuracy and AUC-ROC
Interpretability output	Top features ranked by AGOP diagonal score

2.2.5 XGBoost and LightGBM Configuration

Both models are trained via the scikit-learn API following strong-default settings [Holzmüller et al., 2024]: 1,000 boosting rounds, $\eta = 0.05$, row/column subsample 0.8, and early stopping patience 20 on validation loss. XGBoost uses max depth 6; LightGBM uses `num_leaves=31` to match its leaf-wise growth strategy [Ke et al., 2017].

2.2.6 MLP Configuration

The MLP baseline uses a two-hidden-layer network of sizes $(128, 64)$ with ReLU activations, trained with Adam ($\eta = 0.001$, batch size 256, max 500 epochs) and early stopping with patience 20 on validation loss. All features are z-score standardised; classification outputs use softmax probabilities for AUC computation.

3 Results

This section presents the predictive performance, computational efficiency, scalability, and interpretability comparison between xRFM and three tabular learning baselines: XGBoost, LightGBM, and MLP. All results are generated through the unified experimental pipeline described in Section 2.2.4 and the baseline configuration sections, using the same train/validation/test split and evaluation metrics.

3.1 Predictive Performance

Table 3 reports the final held-out test-set performance for all four models across the five datasets. RMSE is used for regression tasks, while accuracy and AUC-ROC are used for binary classification tasks. Bold formatting indicates the best result for each metric.

Table 3: Test-set performance of all models. **Bold** indicates best result per dataset and metric. RMSE is lower-better; Accuracy and AUC are higher-better.

Dataset	Metric	Model			
		xRFM	XGBoost	LightGBM	MLP
Diamonds	RMSE ↓	1496.72	544.68	**535.99**	734.69
Superconductivity	RMSE ↓	9.454	9.302	**9.209**	11.066
Online Shoppers	Accuracy ↑	0.8897	0.9031	**0.9039**	0.8946
	AUC ↑	0.9018	**0.9306**	0.9306	0.9040
Stroke	Accuracy ↑	0.9413	0.9491	0.9501	**0.9511**
	AUC ↑	0.7605	**0.8388**	0.8231	0.2690
HR Attrition	Accuracy ↑	**0.8673**	0.8571	0.8469	0.8639
	AUC ↑	0.7631	0.7552	**0.7886**	0.7512

Regression tasks. For Diamonds and Superconductivity, tree-based baselines dominate. LightGBM achieves the lowest RMSE on both regression datasets, consistent with previous evidence that boosted trees are strong default models for tabular prediction [Grinsztajn et al., 2022, Holzmüller et al., 2024]. xRFM is close to the boosted-tree baselines on Superconductivity, where its RMSE is 9.454 compared with 9.302 for XGBoost and 9.209 for LightGBM. However, it underperforms substantially on Diamonds, where its RMSE of 1496.72 is much higher than LightGBM's 535.99. This suggests that the AGOP-driven representation used by xRFM is not automatically sufficient for large mixed-type regression tasks without further tuning.

Binary classification tasks. For Online Shoppers and Stroke, XGBoost and LightGBM achieve the strongest AUC-ROC, showing robust ranking performance under class imbalance. xRFM is competitive but not best on these two tasks, reaching AUC values of 0.9018 and 0.7605 respectively. Its most favourable result is HR Attrition, the smallest dataset in the benchmark. On this dataset, xRFM achieves the highest accuracy (0.8673), although LightGBM has the highest AUC (0.7886). This pattern suggests that xRFM can be useful in small-sample settings where its feature-learning mechanism may identify stable directions, but AUC-based ranking performance still favours tree ensembles in several cases.

Overall performance pattern. The main performance conclusion is that XGBoost and LightGBM remain stronger general-purpose models for tabular prediction, while xRFM has selective advantages rather than universal dominance. This is aligned with recent tabular benchmarking work showing that deep or feature-learning approaches can be competitive but require careful adaptation to match boosted-tree baselines [Grinsztajn et al., 2022, Holzmüller et al., 2024, Hollmann et al., 2025].

3.2 Scalability Analysis

To evaluate performance and training cost as a function of data size, we conducted subsampling experiments on Diamonds and Superconductivity. Table 4 reports RMSE and total training time for each sample size. The Diamonds curves are visualised in Figure 1.

Table 4: Scalability experiment: test RMSE and training time (seconds) as a function of training set size. Dash (–) indicates configurations not run.

Dataset	n_{train}	xRFM		XGBoost		LightGBM		MLP	
		RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)
Diamonds	1,000	1657.0	0.77	846.7	0.80	1003.4	0.13	1081.0	2.27
	5,000	1655.9	11.03	628.9	0.39	622.7	0.11	901.1	10.12
	10,000	1577.6	39.31	587.8	0.49	592.3	0.15	888.9	10.33
	30,000	1487.5	226.44	542.7	0.59	535.3	0.29	743.5	22.23
Supercond.	1,000	14.07	0.08	13.76	0.39	13.91	0.15	15.39	2.52
	5,000	11.30	0.47	10.81	1.00	10.71	0.46	13.05	8.58
	10,000	9.81	1.07	9.51	1.73	9.50	1.10	11.22	28.35
	12,757	9.46	1.56	9.28	1.64	9.21	1.04	–	–

Scalability observations. The Diamonds subsampling experiment exposes a severe scalability and accuracy limitation for xRFM. Its training time increases from 0.77s at $n = 1{,}000$ to 226.44s at $n = 30{,}000$, while RMSE only improves from 1657.0 to 1487.5. In contrast, LightGBM reaches 535.3 RMSE at $n = 30{,}000$ while training in less than one second. This indicates that xRFM's gradient-based feature-learning step can become expensive on large mixed-type regression data.

Superconductivity gives a more favourable view of xRFM. As n_{train} increases from 1,000 to 12,757, its RMSE improves from 14.07 to 9.46, approaching XGBoost and LightGBM. Its training time also scales moderately from 0.08s to 1.56s. The contrast between Diamonds and Superconductivity suggests that xRFM may be more competitive when the data is fully numerical and high-dimensional, while it struggles more on mixed categorical/numerical datasets with strong threshold-like patterns.

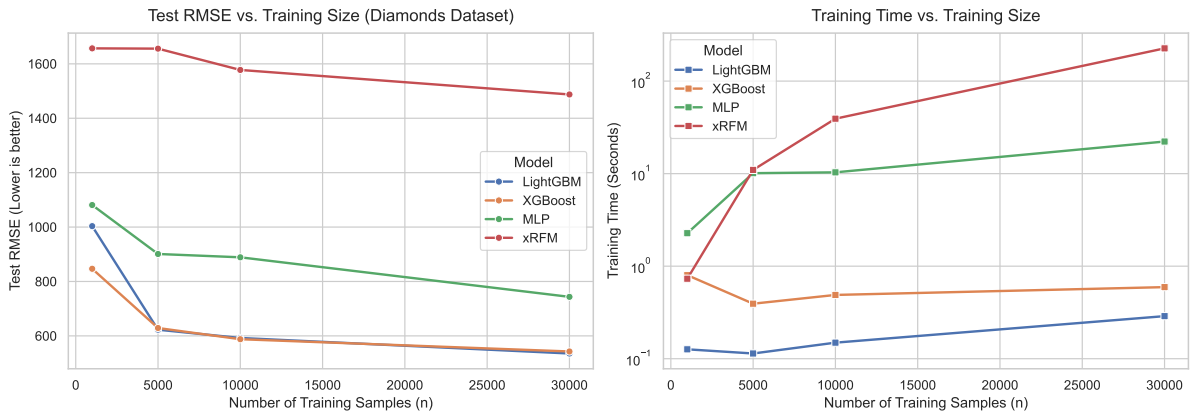

Figure 1: Scalability analysis on the Diamonds dataset, comparing test RMSE (left) and training time (right) across varying training set sizes.

3.3 Interpretability Comparison

The interpretability advantage of xRFM comes from its AGOP-derived feature importance. In our implementation, the feature-importance extraction routine first attempts to read the learned AGOP matrix or diagonal values from the internal xRFM model and then ranks feature names by the resulting values. Table 5 summarises the top-five xRFM features for selected datasets.

Table 5: xRFM AGOP-diagonal feature importance extracted from the trained models.

Dataset	Top AGOP-ranked features (1st–5th)
Diamonds	`y`, `x`, `carat`, `z`, `depth`
Superconductivity	`number_of_elements`, `mean_atomic_mass`, `wtd_mean_atomic_mass`, `gmean_atomic_mass`, `wtd_gmean_atomic_mass`
Online Shoppers	`PageValues`, `ExitRates`, `ProductRelated`, `ProductRelated_Duration`, `BounceRates`
Stroke	`avg_glucose_level`, `age`, `bmi`, `work_type_Self-employed`, `smoking_status_smokes`
HR Attrition	`OverTime_No`, `OverTime_Yes`, `BusinessTravel_Travel_Frequently`, `TotalWorkingYears`, `BusinessTravel_Non-Travel`

The AGOP rankings are broadly interpretable. For Diamonds, xRFM focuses on physical size variables such as x, y, z, and carat, which are plausible drivers of price. For Online Shoppers, `PageValues` is ranked highest, which is consistent with the idea that pages with higher commercial value are strongly related to purchase intention. For Stroke, glucose level, age, and BMI are prioritised, matching common risk factors used in health-related predictive modelling. For HR Attrition, overtime and business-travel variables dominate, suggesting that xRFM identifies work-intensity indicators as major contributors to attrition predictions.

Table 6 compares AGOP diagonal importance with PCA, mutual information, and permutation importance at the method level. We do not treat these explanations as interchangeable: each method answers a different question about the model or data. AGOP asks which features the learned xRFM function is locally sensitive to; PCA asks which directions explain input variance; MI asks which individual features have statistical dependence with the target; and permutation importance asks how much test performance drops when a feature is corrupted.

Table 6: Conceptual comparison of interpretability methods used for tabular features.

Method	Supervised?	Model-specific?	Main interpretation
AGOP diagonal	Yes	Yes	Sensitivity of xRFM to each feature
PCA loadings	No	No	Directions of maximum input variance
Mutual information	Yes	No	Univariate dependence with target
Permutation importance	Yes	Model-agnostic	Performance drop after feature shuffle

Compared with PCA, AGOP is more directly connected to prediction because it is computed from gradients of the learned function rather than from input covariance alone. Compared with MI, AGOP can reflect the behaviour of a multivariate model, whereas MI is typically calculated feature by feature. Compared with permutation importance, AGOP is cheaper to extract after model training, but permutation importance has a clearer link to performance degradation. Therefore, the AGOP diagonal is best interpreted as an internal explanation of xRFM's learned feature geometry rather than a universal ground-truth ranking of feature importance.

4 Discussion

4.1 xRFM: When Feature Learning Helps and When It Does Not

The empirical results partially support the motivation behind xRFM. The AGOP mechanism is designed to learn task-relevant feature directions rather than relying only on the original coordinate system [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. This is most visible in

the HR Attrition dataset, where xRFM achieves the highest accuracy among the four models. Since HR Attrition has only 882 training examples after splitting, this result suggests that xRFM can be competitive in small-data settings where a structured feature-learning prior may help stabilise prediction. The feature-importance ranking for this dataset is also plausible: overtime, business travel, and total working years are all work-condition variables that can reasonably relate to attrition.

However, the results also show clear limitations. The strongest negative case is Diamonds, where xRFM obtains 1496.72 RMSE, compared with 544.68 for XGBoost and 535.99 for LightGBM. This is not a small gap: xRFM is almost three times worse than the best tree-based model on this dataset. A likely algorithm-design explanation is that Diamonds contains strong axis-aligned effects and threshold-like categorical structure, such as cut, colour, clarity, and size variables. Boosted decision trees are highly effective at capturing such split-based interactions [Friedman, 2001, Breiman, 2001, Chen and Guestrin, 2016, Ke et al., 2017]. By contrast, xRFM estimates task-relevant feature geometry through gradients and kernel-style feature learning. This can be less efficient when the dominant predictive structure is already well represented by simple tree splits.

The scalability experiment strengthens this interpretation. On Diamonds, xRFM improves only modestly as n_{train} increases from 1,000 to 30,000, while its training time increases from less than one second to more than 226 seconds. This indicates that simply adding more data does not solve the inductive-bias mismatch for this dataset. In contrast, on Superconductivity, xRFM improves from 14.07 RMSE at 1,000 examples to 9.46 RMSE at the full training size, approaching the boosted-tree baselines. This dataset is fully numerical and high-dimensional, which may better match AGOP's objective of identifying useful continuous directions in feature space.

Therefore, xRFM appears most promising when the data are numerical or moderately high-dimensional, and when interpretability through learned feature sensitivity is important. It is less attractive when the task is large-scale mixed-type tabular regression with strong categorical threshold effects, where XGBoost and LightGBM remain more accurate and substantially more efficient [Grinsztajn et al., 2022, Holzmüller et al., 2024]. Future improvements should focus on dataset-specific hyperparameter tuning, better handling of categorical variables, more efficient AGOP approximation, and integration with tree-style partitioning mechanisms.

4.2 XGBoost and LightGBM: Performance and Efficiency

XGBoost and LightGBM deliver competitive or superior performance across all five datasets while remaining the fastest models to train. On Diamonds, both GBDTs outperform xRFM by a large margin in RMSE (544.68 and 535.99 versus 1496.72), consistent with the strong axis-aligned inductive bias of tree-based splits [Grinsztajn et al., 2022]. On HR Attrition ($n = 882$, $d = 51$), xRFM marginally surpasses XGBoost in accuracy (0.8673 versus 0.8571), suggesting that kernel-style feature learning offers a modest advantage only in a restricted small-data regime. LightGBM is the fastest model at every large Diamonds scale, with near-linear growth in training time (Table 4).

4.3 MLP: Performance and Scalability Limitations

On Diamonds, MLP underperforms both XGBoost and LightGBM by a large margin in RMSE (743.5 versus 542.7 and 535.3 at $n = 30{,}000$), and a similar pattern is observed on Superconductivity (11.22 versus approximately 9.5 RMSE) (Table 3), indicating weaker ability to exploit feature interactions without extensive architecture tuning [Tolstikhin et al., 2021, Holzmüller et al., 2024]. In classification tasks, performance is unstable: while accuracy can be comparable in some cases, such as Stroke, probability ranking degrades significantly, as reflected by the low AUC of 0.269 on Stroke. This highlights sensitivity to class imbalance and calibration issues [Walde

et al., 2004]. Unlike tree-based methods, MLPs rely on global gradient-based optimisation over continuous parameter spaces. This makes them less aligned with the piecewise and often sparse structure of tabular features, resulting in lower data efficiency.

5 Conclusion

In this project, we replicated xRFM, an AGOP-driven feature learning method for tabular data, and conducted a reproducible comparative evaluation against XGBoost, LightGBM, and MLP across five public datasets covering regression and binary classification. A unified pipeline was used for preprocessing, splitting, training, and evaluation, reducing the risk of cross-model discrepancies in data usage and metric calculation.

The experimental findings are threefold. First, tree-based gradient boosting models remain the strongest general-purpose baselines for the evaluated tabular tasks, with LightGBM and XGBoost providing the best overall balance of predictive accuracy and efficiency. Second, xRFM shows selective promise: it is competitive on Superconductivity and achieves the highest accuracy on HR Attrition, suggesting that AGOP-based feature learning can be useful in some small-data or numerical feature-learning settings. However, it underperforms substantially on Diamonds and scales poorly on that dataset, indicating that AGOP feature learning is not automatically superior to tree-based splitting on large mixed-type tabular regression. Third, MLP performance is less stable, especially under class imbalance and large-scale tabular regression.

The interpretability analysis shows that xRFM provides meaningful AGOP-diagonal feature rankings, such as size-related variables for Diamonds, `PageValues` for Online Shoppers, and overtime-related variables for HR Attrition. These rankings are useful as internal explanations of xRFM's learned feature geometry, but they should be interpreted alongside external methods such as permutation importance. Future work should investigate stronger xRFM hyperparameter tuning, more efficient AGOP approximation, improved categorical handling, and broader evaluation on sparse and multi-class tabular datasets.

A AGOP-Based Splitting Criterion: Scratch Implementation

The xRFM tree selects the split direction at each node as the leading eigenvector v_1 of the AGOP $G_f = \mathbb{E}_x[\nabla f(x) \nabla f(x)^\top]$ [Beaglehole et al., 2025]. We implement this criterion from scratch using Ridge regression ($f(x) = w^\top x + b$, $\alpha = 10^{-3}$). For a linear model the gradient is constant so the AGOP simplifies to $G_f = ww^\top$, giving the closed-form diagonal $[G_f]_{ii} = w_i^2$ and split direction $v_1 = w/\|w\|$. The optimal threshold t^* is found by scanning 30 candidate quantiles of $v_1^\top X$ and picking the one that minimises weighted target variance. The full implementation is in `bonus/agop_split.py`.

Verification. We use a synthetic dataset ($n = 300$, $d = 10$) with $y = 3x_0 - 2x_1 + \varepsilon$, so the ground-truth signal features are $\{x_0, x_1\}$. We compare the scratch AGOP diagonal w_i^2 against a numerical AGOP diagonal computed by finite-difference gradients of the fitted xRFM predictor (black-box use only).

Table 7: AGOP diagonal on the synthetic dataset. Noise features x_2–x_9 score ≤ 0.03 (scratch) and ≤ 0.30 (xRFM).

Feature	Scratch w_i^2	xRFM numerical
x_0 (coef. +3)	9.23	13.03
x_1 (coef. −2)	3.93	5.28
x_2–x_9 (noise)	≤ 0.03	≤ 0.30
Top-2 identified	$\{x_0, x_1\}$	$\{x_0, x_1\}$
Cosine similarity	**0.9985**	

Both methods identify $\{x_0, x_1\}$ correctly (overlap 2/2, cosine similarity 0.9985), confirming that the scratch criterion recovers the same split direction as the xRFM library.

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